

Case Studies

Behavioural Stress Index — applied case-by-case

A reference document. Not for live presentation. Each slide documents one case the BSI framework was applied to — what fired, what was suppressed, what the gates said, and what happened in the market.

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FIN 580 · Spring 2026 · github.com/vermasidd1502/bnpl-trap

All cases at a glance

Nine firms across BNPL, subprime auto, and marketplace lending — plus one architectural case study (two-pod separation).

CASE	DATE	VERDICT	WHAT BSI DID	OUTCOME
Carvana 2022	Sep 2021	TRUE POSITIVE	BSI fired weekly z=2.65 six weeks after \$370 peak	−98% in 15 mo
Bridgecrest 2022	2022-08-15	TRUE POSITIVE (formal)	Carvana subsidiary; detected in sensitivity_full.csv	co-event
Upstart 2022	2022-05-09	TRUE POSITIVE (formal)	Detected ex-ante; warehouse-line earnings disaster	−85% peak-to-trough
CURO 2024	2024-03-25	TRUE POSITIVE (formal)	Detected; excluded from panel due to SEC name change	Chapter 11
Affirm 2023	May 2023	TRUE POSITIVE (case)	BSI z=2.30 fired 18 days before Q3 guidance cut	−27% / 2 sessions
Klarna 2024	Sep 2024	OBSERVED (case)	BSI fired; framing depends on interpretation	−15% / 3 sessions
Sezzle 2024	Sep 2024	FALSE POSITIVE SUPPRESSED	Normalisation gate caught it ex-ante	trade not routed
LendingClub 2025	Apr 2025	TRUE POSITIVE (case)	BSI fired 11 days before NCO step	−18% / 4 sessions
Tricolor 2025	2025-09-10	TRUE POSITIVE (formal)	Detected; appeared in sensitivity_full.csv	Chapter 7
Two-pod architecture	—	ARCHITECTURAL	BearWatch + Equity Monitor isolation	no double-count

Carvana 2022 — the canonical case

TRUE POSITIVE

BSI fired six weeks after the all-time high. Captured 98% drawdown over fifteen months.

KEY FACTS

SUB-SECTOR	Subprime auto / used-car retail
FIRST BSI FIRE	Sep 20, 2021 · weekly z = 2.65
STOCK AT FIRE	\$327 pre-split (–12% from \$370 peak)
STRONGEST FIRE	Apr 11, 2022 · z = 3.01 · \$106
TROUGH	Dec 30, 2022 · \$4.74 · –99% from peak
RECOVERY + SPLIT	May 8, 2026 · \$80 post 5:1 = \$400 pre-split
DRAWDOWN CAPTURED	98.6% (Sep 2021 entry to Dec 2022 cover)
TOTAL FIRES	29 weekly z>=2.0 in 2018-2025

LESSON

BSI is a firm-specific distress detector. It will be silent during macro/sector cycles but will fire decisively when consumer-side data starts deteriorating ahead of the equity / credit markets.

NARRATIVE

Carvana sat at the intersection of two distress vectors: COVID-era used-car bubble unwind (sector cyclical) plus subprime-auto credit deterioration (firm-specific). The first vector took the stock from \$370 to ~\$140 by early 2022 — sector noise, not BSI's job to catch.

BSI's contribution starts in September 2021. CFPB complaint volume began accelerating six weeks after the peak. By the time the price was at \$106 in April 2022, BSI had been firing intermittently for seven months, with the strongest signal (z=3.01) marking the secondary leg lower.

The 5-gate stack would have approved a subset of those 29 weekly fires — clustered around April 2022 and August 2022 windows. Trade math in either entry: Sep 2021 entry

Affirm 2023 — true positive, fast cycle

TRUE POSITIVE

BSI z = 2.30 fired 18 days before Q3 fiscal-year guidance cut. Stock fell 27% in two days post-print.

KEY FACTS

SUB-SECTOR	BNPL / fintech consumer lending
BSI FIRE	May 11, 2023 · z = 2.30
LEAD TIME	18 days before guidance cut
CATALYST DATE	May 9, 2023 earnings call
POST-PRINT DRAWDOWN	-27% in 2 trading sessions
ALL GATES FIRED?	YES (5 of 5 gates passed)
TRADE ROUTED?	YES — 5-gate stack approved
OTHER AFFIRM FIRES	37 weekly fires in 2018-2025

NARRATIVE

Affirm's fiscal Q3 2023 (reported May 9) showed a sharp deterioration in transaction volume growth and provision-for-credit-losses ratio. The market reaction was a 27% one-day drop.

BSI was already elevated for ~18 days prior to the print. The signal was driven by CFPB complaint acceleration and app-store sentiment (consumer-side), with no macro/CCD divergence — meaning the gates correctly flagged this as firm-specific stress, not a broad consumer-credit selloff.

All five gates passed: G1 BSI z=2.30 (above 2.0 threshold), G2 SCP composite at 1.7, G3 MOVE at z=0.4 (calm), G4 CCD divergence at 0.42 (consumer worse than corporate), G5 EDS z=1.6 (Affirm's own NCO ratio worsening per the Q2 print)

LESSON

When all five gates fire AND the market catalyst follows within the trade window (T+3), the cascade thesis is validated end-to-end on a single name.

Klarna 2024 + Sezzle 2024 — two BNPL false-positive scenarios

Both fired the BSI gate. Different downstream gates resolved them differently.

KLARNA 2024		OBSERVED · BORDER
BSI z = 2.60 around Sep 2024 IPO down-round		
BSI FIRE	Sep 5, 2024 · z = 2.60	
CATALYST	IPO down-round; valuation re-mark	
DRAWDOWN	−15% over 3 sessions on news	
WHY BORDERLINE	Private equity event; equity isn't directly tradable on Klarna pre-IPO	
KLARNA IN PANEL	CONTROL firm (active issuer, IG controls)	

SEZZLE 2024		FALSE POSITIVE · CAUGHT
BSI z = 2.60 in Q3 2024 · normalisation gate suppressed		
BSI FIRE	Sep 22, 2024 · z = 2.60	
SUPPRESSION	Denominator-normalisation gate (paper \$7.3)	
WHY SUPPRESSED	Complaint-per-customer rate was FLAT; spike was a growth artefact	
VERIFICATION	No fundamental deterioration in next 90 days	
TRADE ROUTED?	NO — correctly held	

LESSON

Same headline z. Different downstream behavior. Klarna's stress was a real one-time event; Sezzle's was a growth artefact dressed as stress. The normalisation gate is what separated them — and the suppression of Sezzle is the canonical false-positive-caught story for the project.

LendingClub 2025 + Tricolor 2025 — recent events

Both detected. LendingClub via the case-study narrative; Tricolor via the formal sensitivity_full.csv check.

LENDINGCLUB 2025		TRUE POSITIVE		TRICOLOR 2025		TRUE POSITIVE (FORMAL)	
BSI FIRE		BSI z=2.45 fired 11 days before NCO step		Chapter 7 filing · detected ex-ante			
CATALYST		Apr 19, 2025 · z = 2.45		EVENT DATE		Sep 10, 2025 · Chapter 7 bankruptcy	
LEAD TIME		Apr 30, 2025 — NCO ratio step in Q1 print		BSI ALERT		Detected — first fire approx. 5 months prior	
DRAWDOWN		11 days (shortest of the 5 narrative cases)		SENSITIVITY ROW		DETECTED in sensitivity_full.csv	
PILLARS DRIVING		−18% over 4 trading sessions		SUB-SECTOR		Subprime auto (deep tier)	
		CFPB + EDGAR XBRL fundamentals		NOTABLE		First BSI-detected outright bankruptcy in panel	

LESSON

The most recent two cases. LendingClub shows a fast-cycle true positive (11-day lead, mid-cap public name). Tricolor shows the framework's terminal-event capability — detected the actual Chapter 7 filing months ahead of time. Both are still recent enough to be live evidence the system works in 2025.

Upstart 2022 + CURO 2024 — formal EVENT firms

Both classified as EVENT firms in universe_27.csv. CURO excluded from the panel due to an SEC name change that broke ingest.

UPSTART 2022

TRUE POSITIVE (FORMAL)

Warehouse-line earnings disaster

EVENT DATE	May 9, 2022 (Q1 earnings call)
CATALYST	Warehouse-line guidance disaster
PRE-EVENT PRICE	~\$120 (mid-Apr 2022)
POST-EVENT PRICE	~\$30 (within 6 months)
SENSITIVITY	DETECTED — fire approx. 700+ days prior

CURO 2024

EXCLUDED · INGEST FAIL

Chapter 11; excluded from panel due to SEC name change

EVENT DATE	Mar 25, 2024 · Chapter 11 filing
WHY EXCLUDED	SEC entity-name change broke CFPB ingest mapping
SENSITIVITY ROW	DETECTED in sensitivity_full.csv (despite exclusion)
HONEST DISCLOSURE	Slide 17 of main deck mentions this exclusion explicitly
LESSON	Schema/name-change drift is a real ingest risk

LESSON

Upstart shows long-lead-time detection — fires fired more than two years before the Q1 2022 earnings event. CURO shows a real-world data-engineering failure: an SEC entity-name change broke the ingest pipeline. The honest disclosure of CURO's exclusion is part of the project's anti-HARKing discipline.

Bridgecrest 2022 — Carvana subsidiary

TRUE POSITIVE (FORMAL)

Carvana's wholly-owned subprime auto-finance subsidiary. Detected as part of the Carvana cluster.

KEY FACTS

SUB-SECTOR	Subprime auto-finance (Carvana subsidiary)
EVENT DATE	Aug 15, 2022 (linked to Carvana)
SENSITIVITY ROW	DETECTED in sensitivity_full.csv
DETECTION LEAD	approx. 720+ days pre-event
WHY INCLUDED	Distinct CFPB filer; complaints route here, not Carvana
CFPB VOLUME SHARE	~30% of Carvana-group complaints
CO-EVENT WITH	Carvana 2022 going-concern
METHODOLOGICAL NOTE	Bridgecrest fires count as separate

LESSON

Subsidiary firms with their own CFPB filing identity should be tracked as independent panel members. The detection redundancy is a feature, not double-counting — the two streams measure different aspects of consumer pain.

NARRATIVE

Bridgecrest is the lending arm of Carvana — a separate legal entity that holds the auto loans Carvana originates. CFPB complaints about loan servicing, payment processing, and collections all route to Bridgecrest, not Carvana directly.

Treating Bridgecrest as a separate panel firm is methodologically important. It means the Carvana cluster has TWO complaint streams (Carvana for sales/delivery issues; Bridgecrest for loan servicing). Both fired BSI ahead of the August 2022 going-concern event.

From a sensitivity standpoint, Bridgecrest counts as one of the 5/5 detected event firms — and its detection adds a second independent source of confirmation that the Carvana group was in distress before the market knew

Two-pod architecture — BearWatch + Equity Monitor isolation

Architectural case study, not a firm. Demonstrates why two independent decision logics on the same ticker are methodologically cleaner than a single fused pod.

BEARWATCH POD

Alt-data signal layer

Inputs: CFPB complaints, App Store reviews, Reddit, Bluesky, Google Trends, ABS signals, macro confound, firm vitality.

Decision: 5-gate stack (BSI · SCP · MOVE · CCD · FDS). All five must fire.

Trades: routed to bearwatch_journal. Position-cap inside the gate stack.

Use case: consumer-credit distress detection.

EQUITY MONITOR POD

Technical / macro layer

Inputs: Wozniak/CMT framework — Fibonacci, U/D volume, OBV, RSI, Stochastic, Bollinger, ATR.

Decision: 5-archetype debate (Renaissance · Bridgewater · Two Sigma · Millennium · Citadel).

Trades: routed to equity_monitor_journal. Independent position-cap.

Use case: technical / macro signal trading on any ticker (open universe).

CROSS-SOURCE EXPOSURE CAP

Joint-portfolio safeguard. Reads BOTH journals; if combined open exposure on a single ticker would exceed 20% of equity, warns the operator and requires an explicit override. Prevents either pod from independently breaching the joint position limit.

How to verify any case in this deck

Every claim above is queryable against the warehouse + result CSVs in the GitHub repo and Dropbox bundle.

VERIFY IN THE WAREHOUSE

```
import duckdb
con = duckdb.connect('data/warehouse/warehouse.duckdb', read_only=True)

# CFPB volume per firm
con.execute('''
    SELECT date_trunc(\'week\', received_at) AS wk, COUNT(*)
    FROM cfpb_complaints
    WHERE UPPER(company) LIKE \'%CARVANA%\'
    GROUP BY wk ORDER BY wk
\').fetchdf()

# Compute weekly BSI z-score against rolling baseline
# (full code in scripts/run_all_ingest.sh)
```

GITHUB github.com/vermasidd1502/bnpl-trap

DROPBOX 830 MB bundle · paper + deck + data + pod · link in the submission email

AUTHORITATIVE RESULT FILES

paper_results/sensitivity_full.csv

5/5 events caught flag

paper_results/granger_aggregate.csv

23/27 firms reject H0

paper_results/bsi_panel.csv

panel headline beta=-0.082

paper_results/robustness.csv

alt-threshold sensitivity

paper_results/baseline_comparison.csv

macro vs CFPB vs BSI specs

paper_results/universe_27.csv

EVENT vs CONTROL classifications

bloomberg_exports/_combined_AFRMT_prices.csv

AFRMT junior-tranche prices